# Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Agentic AI · Multi-Agent Systems · LangGraph · Evaluation Pipelines · Python · React · AWS · Architecture · 9yr Production

+1-757-656-9038 | venkatalakshmigmu@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

## PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience and 2+ years building agentic AI systems, multi-agent orchestration platforms, and evaluation pipelines - exactly the work AWS ProServe's Migration & Modernization Engineering team does. Leasel is the multi-agent proof point: an 8-agent LangGraph StateGraph where AI agents and human consultants collaborate as a unified delivery team - with conditional routing, dynamic task delegation based on confidence, shared long-term memory across agents and sessions, tool invocation, and deterministic HITL guardrails that prevent unsafe decisions on regulated paths. This is the same planner-executor-validator architecture the PCAM platform applies to enterprise migration workflows. AgentProbe is the evaluation pipeline proof point: a production platform with a 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-Judge scoring, pass@k multi-trial harness, CI-integrated regression gate blocking drift before deployment, and MCP integration - the evaluation and observability infrastructure that ensures agentic systems remain reliable at scale. At Entrata, I owned production enterprise systems for 9 years (Python, AWS S3/EC2/RDS/Glue, Docker microservices, RabbitMQ event-driven async, FastAPI, React) at 99.5% uptime - the distributed architecture and operational excellence standards production agentic systems require. I design systems, lead projects across engineers, mentor teammates, write production code daily, and own operational excellence from alarms through runbooks. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## TECHNICAL SKILLS

**Agentic AI & Orchestration:** Python (9yr), LangGraph, LangChain, Multi-Agent Systems, Agent Orchestration, Tool Calling, Function Calling, Multi-Step Workflows, Conditional Routing, Planning & Execution, HITL Guardrails, Generative AI, LLM Fundamentals, Transformer Architecture, Anthropic API, OpenAI API, AWS Bedrock, MCP, Claude Code, Prompt Engineering, Context Engineering

**Evaluation & Platform:** Evaluation Pipelines, LLM-as-Judge, Hallucination Detection, Groundedness, pass@k Harness, CI Regression Gate, RAG Pipelines, ChromaDB, Vector Databases, Semantic Search, FastAPI, React.js, TypeScript, JavaScript, REST APIs, API Design, AWS (S3, EC2, RDS, Glue, Lambda, DynamoDB, Step Functions), Docker, Microservices, GitHub Actions CI/CD

**Engineering Leadership:** Architecture Design, System Design, Distributed Systems, Reliability Engineering, Incident Response & RCA, Observability, Logging, Runbooks, Automated Testing, Performance Engineering, Technical Mentoring (5+ engineers, 600+ students), Code Reviews (300+), Design Docs & RFCs, PHI/HIPAA Compliance, PostgreSQL, MySQL, Oracle, Redis, RabbitMQ, Linux, Git, Agile/Scrum

## PROJECTS

### Leasel: Multi-Agent Orchestration System - AI + Human Unified Delivery

Multi-Agent · LangGraph · HITL · Orchestration · Production

Python · LangGraph (8-agent) · FastAPI · RAG · SQLite · MCP · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production multi-agent system where AI agents and human reviewers collaborate as a unified delivery team - the same architecture the PCAM platform applies to enterprise migration orchestration. **Agent orchestration:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with conditional routing and dynamic task delegation based on confidence - planner-executor-validator pattern. **Shared context:** SQLite knowledge repository persisted across all agents and sessions - the coordination layer enabling parallel agent and human workflows. **Tool invocation:** agents invoke retrieval, structured data, and configuration tools with grounded, citation-backed outputs. **Guardrails:** deterministic HITL escalation blocks unsafe decisions. 93.6% precision, 87.9% F1 across 1,240 test cases.

### AgentProbe: Production Evaluation Pipeline - Non-Deterministic AI Systems

Evaluation Pipeline · LLMOps · CI Gate · MCP · Open Source

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · AWS Bedrock-compatible · Anthropic API · OpenAI API |
github.com/venkatalakshmiparimala/AgentProbe

Production evaluation platform ensuring agentic systems remain reliable at scale - the observability and quality infrastructure that enterprise migration platforms require. **Evaluation pipeline:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness surfacing variance in non-deterministic outputs. **CI regression gate:** GitHub Actions integration blocks quality drift before deployment - making evaluation a first-class engineering practice. **Platform:** multi-provider LLM routing (Anthropic, OpenAI, Ollama, Bedrock-compatible); MCP integration; 15-endpoint FastAPI backend. Zero regressions across 1,240+ test cases. Open-source, enterprise-deployable.

### AI on FHIR: Full-Stack Agentic System - React, APIs & AWS

Full Stack · React · AWS · Tool Use · HIPAA

Python · Flask · React.js · TypeScript · Docker · AWS · BERT · spaCy · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack agentic system with React frontend, Python backend, and AWS infrastructure - LLM-powered tool invocation translates natural-language queries into structured API calls; HIPAA-compliant RBAC with authentication and full data isolation; complete audit trail; Docker containerization; ~60% reduction in manual retrieval time. Demonstrates the full-stack ownership (React through backend through cloud infrastructure) and secure enterprise integration pattern this role requires.

## EXPERIENCE

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer - Production Agentic AI Delivery**

- Designed and deployed a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - sole technical owner from requirements through architecture, implementation, evaluation pipeline, and production deployment in weeks; wrote design docs, built HITL guardrails and validation gates for regulated decisions; delivered runbooks for ongoing operation.

### George Mason University

Aug 2024 - May 2026

**Graduate Teaching Assistant & Research Associate**

- Built ML pipelines (PyTorch, scikit-learn, $R^2$=0.977); designed observability dashboards; mentored 600+ students in Python, SQL, and distributed systems; delivered complex technical outputs for non-technical stakeholders.

### Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer · Lehi, UT (Remote)**

- **Architecture ownership and system design** - designed and led Docker microservices architecture redesign (50% scalability improvement, 30% deployment time reduction); RabbitMQ event-driven async system with retry logic, idempotency guarantees, and dead-letter handling; AWS S3 document migration (100% PHI compliance, zero data loss); owned HUD compliance pipeline (180,000 domain records, 100K+ housing units) for 7 years at 99.5% uptime - full design-through-operations ownership.
- **Production engineering and operational excellence** - Python, FastAPI/Flask REST APIs, React.js, TypeScript, AWS S3/EC2/RDS/Glue, Docker, PostgreSQL/MySQL/Oracle, Redis; 200+ SQL optimizations (30% performance gain); internal debugging tools reducing triage time 40%; GitHub Actions CI/CD with automated testing; incident response and RCA; meaningful alarms and runbooks; bug count below 10 for 6+ consecutive months; 20+ production modules across 120+ Agile sprints.
- **Technical leadership and mentoring** - led projects spanning multiple engineers; wrote design docs and architecture decisions; mentored 5+ engineers on system design, agentic AI patterns, and production practices; 300+ peer code reviews; cross-functional delivery with product, compliance, and operations stakeholders; Python/Bash automation eliminating 80% of manual workload.

### Entrata India - Pune

Jan 2017 - Sep 2018

**Associate Software Engineer → Trainee**

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

## EDUCATION

**M.S. Data Analytics Engineering -** George Mason University, Virginia — May 2026 · GPA 3.87

**B.Tech Computer Science & Engineering -** JNTU Anantapur — 2013-2017